Hookworms, Malaria and Vitamin A Deficiency Contribute to Anemia and Iron Deficiency among Pregnant Women in the Plains of Nepal[1]

Michele L. Dreyfuss,*,[2] Rebecca J. Stoltzfus,* Jaya B. Shrestha,† Elizabeth K. Pradhan,* Steven C. LeClerq,* Subarna K. Khatry,† Sharada R. Shrestha,† Joanne Katz,* Marco Albonico‡ and Keith P. West, Jr.*

*Department of International Health, The Johns Hopkins School of Hygiene and Public Health, Baltimore, MD 21205; †Nepal Nutrition Intervention Project-Sarlahi (NNIPS), Nepal Netra Jyoti Sangh, Nepal Eye Hospital Complex, Tripureswor, Kathmandu, Nepal and ‡Schistosomiasis and Intestinal Parasites Unit, Division of Control of Tropical Diseases, World Health Organization, Geneva 27, Switzerland

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

ABSTRACT Anemia and iron deficiency during pregnancy are prevalent in developing countries, but their causes are not always known. We assessed the prevalence and severity of anemia and iron deficiency and their association with helminths, malaria and vitamin A deficiency in a community-based sample of 336 pregnant women in the plains of Nepal. Hemoglobin, erythrocyte protoporphyrin (EP) and serum ferritin were assessed in venous blood samples. Overall, 72.6% of women were anemic (hemoglobin $<$ 110 g/L), 19.9% had moderate to severe anemia (hemoglobin $<$ 90 g/L) and 80.6% had iron deficiency (EP $>$ 70 μmol/mol heme or serum ferritin $<$ 10 μg/L). Eighty-eight percent of cases of anemia were associated with iron deficiency. More than half of the women (54.2%) had a low serum retinol concentration ($<$1.05 μmol/L), 74.2% were infected with hookworms and 19.8% had *Plasmodium vivax* malaria parasitemia. Hemoglobin, EP and serum ferritin concentrations were significantly worse and the prevalence of anemia, elevated EP and low serum ferritin was increased with increasing intensity of hookworm infection. Hookworm infection intensity was the strongest predictor of iron status, especially of depleted iron stores. Low serum retinol was most strongly associated with mild anemia, whereas *P. vivax* malaria and hookworm infection intensity were stronger predictors of moderate to severe anemia. These findings reinforce the need for programs to consider reducing the prevalence of hookworm, malaria infection and vitamin A deficiency where indicated, in addition to providing iron supplements to effectively control anemia. J. Nutr. 130: 2527–2536, 2000.

KEY WORDS: • *anemia* • *iron deficiency* • *pregnancy* • *Nepal*

Iron deficiency is the most common form of malnutrition worldwide and is estimated to affect from 1.3 to 2.2 billion persons (United Nations 1990, World Health Organization 1992). When iron deficiency is sufficiently severe, red blood cell synthesis becomes impaired, and anemia results. Approximately 50% of women and children in less developed countries are anemic (DeMaeyer and Adiels-Tegman 1985), and 60% of anemic women in the world reside in South Asia (ACC/SCN 1992). Globally, the most common cause of anemia is believed to be iron deficiency due to inadequate dietary iron intake, physiologic demands of pregnancy and rapid growth and iron losses due to parasitic infections. However, iron deficiency is not the only cause of anemia. Other prevalent causes of anemia include malaria, chronic infections and nutritional deficiencies of vitamin A, folate and vitamin B-12. The relative contributions of these causes of anemia and iron deficiency vary by sex, age and population and are not well described in many populations.

During pregnancy, iron requirements exceed storage iron for most women (Bothwell and Charlton 1984). The increased need by the body for iron is due to increases in the red cell mass, iron needs of the fetus and iron losses during delivery (Bothwell and Charlton 1984). Although hemodilution from expansion of the plasma volume leads to a "physiologic pregnancy anemia" (DeLeeuw et al. 1966), inadequate iron supply can limit red cell mass expansion and lead to further deterioration in iron status during pregnancy (Viteri 1994) that may pose risks for the pregnant woman and her infant (Allen 1997). Severe anemia during pregnancy is associated with a woman's increased risk of death (Llewellyn-Jones 1965), and moderate to severe anemia is associated with an increased risk of low birth weight (Garn et al. 1981, Murphy et al. 1986) and preterm delivery (Klebanoff et al. 1991, Scholl et al. 1992, Zhou et al. 1998). Iron deficiency and anemia during pregnancy are associated with lower iron stores in the fetus, which

[1] Funded through cooperative agreement HRN-A-00-97-00015-00 between The Johns Hopkins University and the Office of Health and Nutrition, U.S. Agency for International Development. This study was conducted as a collaboration between the Center for Human Nutrition and the Sight and Life Institute in the Department of International Health at The Johns Hopkins School of Hygiene and Public Health and the National Society for Comprehensive Eye Care (Nepal Netra Jyoti Sangh), Kathmandu, Nepal.

[2] To whom correspondence should be addressed at Department of Nutrition, Harvard School of Public Health, 665 Huntington Avenue, Boston, MA 02115.

0022-3166/00 $3.00 © 2000 American Society for Nutritional Sciences.
Manuscript received 14 February 2000. Initial review completed 28 April 2000. Revision accepted 27 June 2000.

may result in iron deficiency anemia (Agarwal et al. 1983, Kaneshige 1981, MacPhail et al. 1980, Milman et al. 1987, Puolakka et al. 1980). In several studies, iron supplementation during pregnancy resulted in greater iron stores in young infants (DeBenaze et al. 1989, Milman et al. 1994, Preziosi et al. 1997).

We assessed a cohort of pregnant women in the rural plains of Nepal to determine the prevalence, severity and infectious and nutritional causes of anemia and iron deficiency. Our goal was to estimate the relative contributions of several causes of anemia and iron deficiency in this population to provide a basis for more effective prevention and control.

SUBJECTS AND METHODS

Study population. The study was conducted in Sarlahi District in the east-central plains (*terai*) of Nepal bordering India. Most of the population is involved in subsistence agriculture, sanitation is very poor, health care services are not widely available and protein–energy and micronutrient malnutrition is prevalent among adults and children (Christian et al. 1998b, West et al. 1991 and 1999). *Ancylostoma duodenale* (hookworm) and *Ascaris lumbricoides* are two species of geohelminths endemic in the area (Navitsky et al. 1998). Historically, malaria is hyperendemic in the terai. An aggressive control program during the 1960s and early 1970s reduced the incidence of malaria to very low levels (Shrestha et al. 1988), but the incidence increased again during the early 1980s. *Plasmodium vivax* is the more common species of malaria parasite in Nepal, but *P. falciparum* is also present (Nepal Malaria Eradication Organization 1986).

The study population consisted of women 12–45 y old who were participating in a randomized community intervention trial of the effect of vitamin A or β-carotene supplementation in women of childbearing age on maternal, fetal and early infant mortality and morbidity rates (West et al. 1999). All women from a 10% subsample composed of 27 of 270 participating wards (local administrative units) in three subdistricts who reported being pregnant during weekly home interviews were enrolled in the main trial and invited to a local clinic, after verbal consent was obtained, for a health and nutrition examination conducted by trained study staff. The three subdistricts were centrally located in the study area and represented the general characteristics of the larger trial area.

This analysis includes data collected from August 1994 through March 1997 from women allocated to placebo and living in the clinic substudy area. Of 621 eligible pregnant women, 388 (62%) visited the clinic; of the 388 women, 368 (95%) had pregnancies confirmed with a β-human chorionic gonadotropin urine test. Twenty-four women visited the clinic for two different pregnancies, but only their first pregnancies were used for analysis, leaving 344 pregnancies. Among these women, 336 (98%) provided a blood specimen for assessment of iron status, and this is the final sample size for this analysis. Clinic attendance was affected by women's refusal to participate and extended absences from home due to a common practice of women returning to their parents' home during pregnancy. Also, women who reported a miscarriage, stillbirth or live birth between the time of pregnancy ascertainment and their scheduled clinic visit were not enrolled in the clinic study. Women who did not enroll were similar in age (24.5 versus 24.4 y, $P = 0.91$) and nutritional status [mid-upper arm circumference (MUAC)[3] 21.3 versus 21.2 cm, $P = 0.61$) to those who did enroll in the study. However, nonparticipants were more likely to be <20 y old (25.5% versus 18.9%, $P = 0.05$), possibly because women are most likely to return to their parents' home for a first pregnancy.

Assessment of nutrition and health status. Age, report of last menstrual period (LMP) and a pregnancy history were collected during the initial home interview for enrollment of pregnancies into the supplementation trial. Date of LMP was based on a combination of prospectively reported menstrual histories and LMP recall. Data on socioeconomic status, including literacy and household possessions, were obtained at a second interview conducted later in pregnancy.

Anthropometric measurements were obtained during the clinic visit. Weight was measured to the nearest 0.1 kg with a battery-powered digital scale (Seca, Columbia, MD). Height was measured to the nearest 0.1 cm with a stationary height board fastened to the clinic wall. MUAC was measured to the nearest 0.1 cm at the midpoint of the left arm with an insertion tape (Zerfas 1975). Triceps and subscapular skinfolds were measured to the nearest 0.2 mm with skinfold calipers (Holtain; Seritex, Carlstadt, NJ). The median of three measurements was recorded for each measure, except for weight, which was measured once.

Iron status was assessed with hemoglobin, erythrocyte protoporphyrin (EP) and serum ferritin concentrations, and vitamin A status was assessed with serum retinol concentration. Blood was collected via venipuncture. Hemoglobin was measured with a Hemocue hemoglobinometer (Mission Viejo, CA), and EP was measured with a hematofluorometer (AVIV Biomedical, Lakewood, NJ). Blood samples were centrifuged at 1530 × *g* for 10 min at room temperature, and serum was collected in 1-mL cryotubes. Serum was immediately stored in liquid nitrogen freezers until transported to Baltimore, where they were stored at −70°C until analysis. Serum ferritin was assessed with a fluorometric immunoassay (Delfia System; Wallac, Gaithersburg, MD). The assay within-day and between-day coefficients of variation were 7.9 and 11.5%, respectively. Serum retinol was determined with reverse phase, isocratic high performance liquid chromatography (Craft 1996), and the assay within-day and between-day coefficients of variation were 2.3 and 3.0–5.7%, respectively.

To detect malaria parasitemia, a thick blood film and a thin blood film were collected, fixed and stained with Giemsa. Blood films were not available for 31 women because they visited the clinic before start-up of the protocol for malaria assessment. An additional 17 women had unreadable blood films, leaving a total of 288 (94% of available specimens) women with blood films available for the detection of malaria. Malaria parasites were counted as a ratio to leukocytes. If <10 parasites were seen after 200 leukocytes were counted, then 500 leukocytes were counted. At least 100 microscope fields were examined in all blood films. The calculation of parasite density was based on 8000 leukocytes/μL of blood (World Health Organization 1991). Malaria species were identified from thick and thin blood films; all infections were identified as *P. vivax*. All specimens identified as positive for malaria parasites were later reread by an experienced malariologist or by another microscopist under his supervision. Only those specimens confirmed positive in the second reading were considered positive in these analyses. A systematic random 10% subsample was reread by the malariologist. Agreement was moderate for the presence of malaria parasitemia (percent agreement = 81%, $\kappa = 0.47$).

For assessment of helminth infections, women were asked to collect a stool specimen in provided containers the evening before or the morning of their clinic visit. Thirty-two of the 336 study subjects have no helminth data because they visited the clinic before start-up of the protocol for helminth assessment. Among the remaining 304 women, 190 (62% of available specimens) returned a stool sample. The Kato-Katz method was used to stain the samples on the day of the clinic visit, and they were read within 1 h of staining (World Health Organization 1994). Specimens were examined by an investigator (M.L.D.) or one of two trained microscopists for the presence of hookworm, *A. lumbricoides* and *Trichuris trichiura* eggs. A subsample of specimens ($n = 71$) were reread by M.L.D. for quality control purposes. Agreement between egg counts in categories of 1000 eggs/g of feces was very good for *A. lumbricoides* (percent agreement = 87%, $\kappa = 0.83$) and good for hookworm (percent agreement = 77%, $\kappa = 0.60$). *T. trichiura* infection was not prevalent enough to estimate agreement.

Women with a hemoglobin concentration of <70 g/L were given a 30-d course of ferrous fumarate capsules containing 120 mg of elemental iron each. *P. vivax* malaria infection was treated with 600 mg of chloroquine on the 1st d, followed by 300 mg/d for the following 3 d. All women found to have helminth eggs in their stool sample at the pregnancy clinic visit were treated with a single 400-mg dose of albendazole when they returned for a second clinic visit 3 mo

[3] Abbreviations used: MUAC, mid-upper arm circumference; AOR, adjusted odds ratio; CI, confidence interval; EP, erythrocyte protoporphyrin; LMP, last menstrual period.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

postpartum. However, women with a hemoglobin concentration of <70 g/L who were also infected with hookworms were immediately given anthelminthic treatment if they were in the second or third trimester of pregnancy.

The study protocol was reviewed and approved by the Nepal Health Research Council in Kathmandu, Nepal, and the Committee on Human Research at the Johns Hopkins School of Hygiene and Public Health in Baltimore, MD.

Data analysis. Our analytic approach to the iron status data involved descriptions of anemia and iron deficiency and their causes. First, we estimated the prevalence and severity of anemia, defined by hemoglobin concentration, and of iron deficiency, defined by serum ferritin and EP concentrations. Second, iron deficiency was examined as a cause of anemia. Finally, we investigated other risk factors as causes of anemia and iron deficiency.

Multiple hemoglobin cutoffs were examined to explore the possibility that some risk factors might be associated with milder anemia and others might be associated with more severe anemia. This is an important possibility because the relationship of anemia to health outcomes depends on the severity of anemia. Anemia was defined as hemoglobin of <110 g/L, and moderate to severe anemia was defined as hemoglobin of <90 g/L (World Health Organization, UNICEF and UNU 1998). We defined severe anemia as hemoglobin of <80 g/L because only 14 women (4.2%) had values below the more conventional cutoff of 70 g/L.

Serum ferritin and EP concentrations were skewed to high values and were log-transformed for analysis. Serum ferritin concentrations were extremely low in this population, and therefore we chose a relatively low cutoff of 10 μg/L to define depleted iron stores (Levin et al. 1993, Romslo et al. 1983). Iron-deficient erythropoiesis was defined as EP of >70 μmol/mol heme (World Health Organization, UNICEF and UNU 1998). Iron deficiency was defined as either serum ferritin of <10 μg/L or EP of >70 μmol/mol heme, and iron deficiency anemia was defined as the presence of iron deficiency with hemoglobin of <110 g/L.

Low serum retinol was defined as <1.05 μmol/L. Parasitic infections were categorized according to their severity and relationship to iron status. Parasite densities for *P. vivax* malaria and *T. trichiura* were uniformly low, so these data are presented as present or absent. For *A. lumbricoides* and hookworm infections, standard cutoffs were used to characterize the population. However, because no relationship was found between *A. lumbricoides* worm burden and iron status, data on this infection were dichotomized for multivariate analyses. Hookworm egg counts were linearly related to all three iron status indicators, so data were analyzed in increasing 1000 eggs/g feces categories for multivariate analyses.

For dichotomous risk factors, differences in hemoglobin, EP and serum ferritin concentrations were compared by Student's *t* test, and differences in anemia and iron deficiency were compared by the χ^2 test. Fisher's exact test was used instead of the χ^2 test in cases where the number of subjects in two or more categories was less than five. Linear trends for continuous and categorical iron status variables were tested by linear regression and by the χ^2 test for trend, respectively. To investigate iron deficiency as a risk factor for anemia, relative risks with 95% confidence intervals (CI) were calculated for indicators of iron deficiency with prevalence rather than incidence data (Kahn and Sempos 1989). Statistical significance was defined as a *P*-value of <0.05.

Adjusted odds ratios (AOR) and 95% CI for anemia, iron-deficient erythropoiesis and depleted iron stores were estimated from logistic regression models, and adjusted mean differences in hemoglobin, EP, and serum ferritin concentrations were estimated from linear regression models. All regression models included variables for intensity of hookworm infection, *P. vivax* malaria parasitemia and low serum retinol regardless of statistical significance. Trimester of pregnancy was retained in all models because iron status and serum retinol were strongly associated with gestational age. Socioeconomic, demographic, anthropometric and other parasitologic variables were retained in models only if statistically significant ($P < 0.05$). Interactions among hookworm infection, *P. vivax* malaria parasitemia and low serum retinol were investigated by stratified bivariate analyses and by inclusion of interaction terms in multivariate regression models. Interaction terms were retained in models if their *P*-value was ≤0.15.

To estimate the maximal proportion of anemia in the population that might be prevented by the elimination of a risk factor, we calculated the attributable fractions of all anemia and moderate to severe anemia for each risk factor. Attributable fraction is the same as attributable risk (Kahn and Sempos 1989) except prevalence ratios are used instead of risk ratios. Adjusted prevalence ratios were calculated from a conversion formula using AOR (Osborn and Cattaruzza 1995). These were then used to calculate attributable fractions with a formula that produces valid estimates when adjusted prevalence ratios are used (Kleinbaum et al. 1982, Rockhill et al. 1998). Data were analyzed using SAS software (SAS Institute, Cary, NC).

TABLE 1

Characteristics of study sample of Nepalese pregnant women[1]

	Frequency
	%
Age, *y* (*n* = 334)	
<20	19.5
20–29	64.0
≥30	16.5
Socioeconomic status (*n* = 316)	
Literacy	12.0
Radio owned	29.1
Activities during the past 7 d	
Smoked cigarettes (*n* = 330)	18.5
Drank alcohol (*n* = 329)	1.5
Parity (*n* = 342)	
0	21.3
1–2	42.5
3–4	23.9
≥5	12.3
Gestational age, *wk*	
≤12	15.5
13–24	63.7
≥25	20.8
Height (<145.0 cm)	17.0
Mid-upper arm circumference (<21.0 cm)	20.2
Serum retinol, *μmol/L* (*n* = 334)	
<0.70	20.7
0.70–1.04	33.5
1.05–1.39	32.3
≥1.40	13.5
Iron supplement use in the past month (*n* = 331)	0.3
Hookworm infection (*n* = 190)	
Negative (0 epg[2])	25.8
Light (1–1999 epg)	61.6
Moderate (2000–3999 epg)	6.3
Heavy (≥4000 epg)	6.3
Ascaris lumbricoides infection (*n* = 190)	
Negative (0 epg)	41.1
Light (1–9999 epg)	53.7
Moderate (10,000–49,999 epg)	4.7
Heavy (≥50,000 epg)	0.5
Trichuris trichiura infection (*n* = 190)	5.3
Plasmodium vivax malaria parasitemia (*n* = 288)	19.8

[1] *n* = 336 except where indicated.
[2] epg, eggs/g of feces.

RESULTS

Characteristics of study sample. Women ranged in age from 15 to 40 y, with 64% between 20 and 29 y (**Table 1**). Few women were literate, and only 29% came from a household in which a radio was owned. Approximately one fifth of the women smoked cigarettes, but alcohol consumption was rare.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

TABLE 2

Indicators of anemia and iron deficiency in Nepalese women by trimester of pregnancy

Trimester of pregnancy (*n*)	Hemoglobin, *g/L* <110	<90	<80	Means ± SD	Erythrocyte protoporphyrin,[1] *μmol/mol heme* >70	Means[2] (−1 SD, +1 SD)	Serum ferritin,[1] *μg/L* <10	Means,[2] (−1 SD, +1 SD)
	%				%		%	
First (52)	63.5	9.6	3.8	104 ± 13	65.4	83 (48, 142)	42.3	11.3 (4.6, 27.7)
Second (214)	72.4	16.8	6.1	101 ± 15	60.6	82 (52, 129)	55.6	9.3 (4.2, 20.4)
Third (70)	80.0[3]	37.1[4]	14.3[3]	94 ± 18[5]	82.9[6]	109 (65, 186)[7]	79.7[4]	6.4 (3.2, 12.7)[5]
Total (336)	72.6	19.9	7.4	100 ± 16	66.0	87 (53, 143)	58.5	8.9 (4.0, 19.8)

[1] $n = 335$.
[2] Geometric means.
[3] χ^2 for trend by trimester of pregnancy, $P < 0.05$.
[4] χ^2 for trend by trimester of pregnancy, $P < 0.005$.
[5] Test for linear trend, $P < 0.0005$.
[6] Significantly different from first and second trimesters combined, χ^2 test, $P < 0.005$.
[7] Significantly different from first and second trimesters combined, Student's *t* test, $P < 0.0001$.

Twenty-one percent of the women were nulliparous, and 36% had given birth to three or more children. Two thirds of the women visited the clinic during the second trimester of pregnancy (13–24 wk) with the other third split between the first and third trimesters. The women were stunted and thin. The height and MUAC of the study sample were 149.9 ± 5.4 and 22.3 ± 1.7 cm, respectively. The vitamin A status of women was poor, with more than half having a serum retinol concentration of <1.05 μmol/L. Only one woman reported the consumption of iron supplements during the past month.

Parasitic infections were extremely common. The prevalence of hookworm and *A. lumbricoides* infections was 74.2 and 58.9%, respectively, and 88.9% of the women were infected with at least one of the three helminths assessed (Table 1). *T. trichiura* infection was rare (5.3%). *P. vivax* malaria parasitemia was present in 19.8% of women, was uniformly low and was not associated with other infections, and its prevalence did not differ by nulliparity (20.3 versus 19.8%, $P = 0.93$). The prevalence of parasitic infections did not vary significantly by trimester of pregnancy.

Anemia was prevalent, and the iron status of the study subjects was poor. Seventy-three percent of the women were anemic, with 19.9% having moderate to severe and 7.4% having severe anemia (**Table 2**). Iron-deficient erythropoiesis was present in 66.0% of the women, and 58.5% had depleted iron stores. The prevalence and severity of anemia and iron deficiency were progressively greater in women examined later in pregnancy (Table 2). Of women examined during the third trimester, 14.3% were severely anemic and 79.7% had depleted iron stores.

Relation of anemia to iron deficiency. The overall prevalence of iron deficiency was 80.6%; 64.0% had iron deficiency anemia, which accounted for 88% of anemia in this population. The relative risk of anemia associated with elevated EP was 1.41 (95% CI 1.19–1.68), and that associated with low serum ferritin was 1.50 (1.28–1.76). When iron deficiency was classified by either or both EP and serum ferritin, there was an increasing linear trend in the prevalence of anemia and of moderate to severe anemia with increasing severity of iron deficiency (**Fig. 1**). These data indicate that iron deficiency was strongly associated with both mild and moderate to severe anemia in this population.

Relation of maternal characteristics to anemia and iron deficiency. Nulliparous women had a lower prevalence of anemia (62.0 versus 75.7%, $P < 0.05$), but the prevalence of moderate to severe anemia did not differ by parity (19.7 versus 20.2%, $P = 0.94$). The prevalence of anemia appeared higher among women ≥ 20 y old than among women < 20 y old (74.7 versus 63.1%, $P = 0.06$). Anemia was not associated with maternal weight, height or MUAC, but women with moderate to severe anemia had significantly smaller triceps and subscapular skinfold measurements (triceps 7.9 ± 2.6 versus 8.7 ± 2.5 mm, $P < 0.05$; subscapular 11.4 ± 3.9 versus 12.5 ± 3.6 mm, $P < 0.05$). Depleted iron stores were associated with significantly higher maternal weight (44.1 ± 5.4 versus 42.6 ± 5.1 kg, $P < 0.05$). Among socioeconomic characteristics, both literacy and radio ownership were inversely associated with anemia but not with indicators of iron deficiency (data not shown).

Risk factors for anemia and iron deficiency: unadjusted analyses. In bivariate analyses, hookworm infection was the most important contributor to anemia and iron deficiency in this population. There was a strong linear trend toward worse values for all three iron status indicators by hookworm inten-

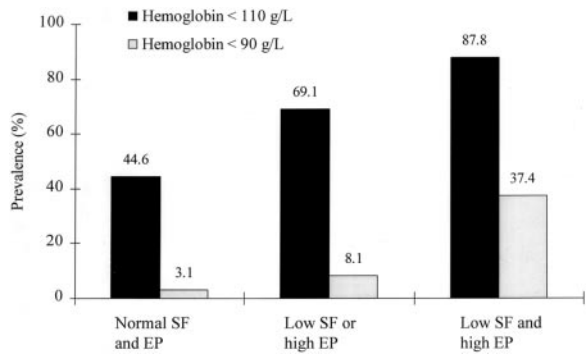

FIGURE 1 Prevalence of anemia in Nepalese pregnant women by degree of abnormal iron status. Cutoffs for indicators of iron status are erythrocyte protoporphyrin (EP) of >70 μmol/mol heme and serum ferritin (SF) of <10 μg/L. For both hemoglobin cutoffs, there is a significant linear trend in anemia prevalence rates by severity of iron deficiency, $P < 0.00001$ by χ^2 test for trend.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

TABLE 3

Indicators of anemia and iron deficiency in Nepalese pregnant women by risk factor[1]

Risk factor	Hemoglobin, *g/L*				Erythrocyte protoporphyrin, *μmol/mol heme*		Serum ferritin, *μg/L*	
	<110	<90	<80	Means ± SD	>70	Means,[2] (−1 SD, +1 SD)	<10	Means,[2] (−1 SD, +1 SD)
	%				%		%	
Hookworm infection								
0 epg[3] (*n* = 49)	55.1	8.2	2.0	106 ± 11	55.1	75 (47, 119)	40.8	12.1 (5.4, 27.2)
1–999 epg (*n* = 88)	68.2	12.5	2.3	104 ± 12	56.3	75 (51, 111)	60.2	9.2 (4.2, 20.0)
1000–1999 epg (*n* = 29)	75.9	24.1	3.4	99 ± 14	72.4	86 (54, 139)	65.5	7.5 (3.7, 15.1)
≥2000 epg (*n* = 24)	75.0[4]	33.3[5]	25.0[5]	90 ± 22[6]	75.0[4]	96 (55, 169)[7]	79.2[5]	7.1 (3.9, 12.9)[8]
Ascaris lumbricoides infection								
Negative (*n* = 78)	66.7	15.4	6.4	102 ± 15	59.0	78 (50, 124)	62.8	8.8 (3.8, 20.6)
Positive (*n* = 112)	67.0	16.1	4.5	102 ± 14	62.2	80 (51, 125)	55.4	9.6 (4.7, 19.5)
Trichuris trichiura infection								
Negative (*n* = 180)	66.7	16.1	5.6	102 ± 15	60.9	80 (51, 125)	58.9	9.2 (4.3, 19.8)
Positive (*n* = 10)	70.0	10.0	0	104 ± 10	60.0	72 (42, 123)	50.0	9.8 (3.8, 24.9)
Plasmodium vivax malaria parasitemia								
Negative (*n* = 231)	68.4	15.6	4.8	102 ± 14	63.5	84 (52, 134)	55.8	9.5 (4.3, 20.8)
Positive (*n* = 57)	77.2[9]	28.1[10]	12.3[10]	97 ± 18	66.7	90 (54, 150)	68.4	7.5 (3.7, 15.0)[11]
Serum retinol								
≥1.05 μmol/L (*n* = 153)	63.4	12.4	4.6	104 ± 13	57.2	77 (50, 119)	52.3	9.6 (4.3, 21.3)
<1.05 μmol/L (*n* = 181)	80.7[9]	26.0[9]	9.4	96 ± 16[12]	72.9[9]	95 (57, 159)[12]	64.1[10]	8.2 (3.7, 18.4)

[1] Means and proportions are unadjusted.
[2] Geometric means.
[3] epg, eggs/g of feces.
[4] χ^2 test for linear trend, $P < 0.05$.
[5] χ^2 test for linear trend, $P < 0.005$.
[6] Significant linear trend, $P < 0.0005$.
[7] Significant linear trend, $P < 0.05$.
[8] Significant linear trend, $P < 0.005$.
[9] $P < 0.005$ by χ^2 test.
[10] $P < 0.05$ by χ^2 test.
[11] $P < 0.05$ by Student's *t* test.
[12] $P < 0.0005$ by Student's *t* test.

sity of infection (**Table 3**). For example, hemoglobin concentration decreased from 106 g/L among uninfected women to 90 g/L among women with moderate to severe infection ($P < 0.0005$). Hookworm infection intensity was strongly associated with anemia at various levels of severity. At ≥2000 eggs/g of feces, the prevalence of moderate to severe anemia was four times that of uninfected women, and severe anemia was 12 times more prevalent. The prevalence of elevated EP and of low serum ferritin concentrations also increased with increasing hookworm infection intensity. Neither *A. lumbricoides* nor *T. trichuris* infection was related to any iron status indicator.

P. vivax malaria parasitemia was associated with anemia, especially more severe anemia (Table 3). There also was evidence that malaria was associated with iron deficiency. Serum ferritin concentration was lower ($P < 0.05$) among women with *P. vivax* malaria parasitemia, and the proportion of women with low serum ferritin tended to be higher.

Low serum retinol concentration was strongly associated with all three iron status indicators in this cohort (Table 3). Women with low serum retinol were more likely to be anemic, to have iron-deficient erythropoiesis and to be iron depleted.

Risk factors for anemia and iron deficiency: multivariate analyses. AOR for anemia and iron deficiency, calculated from multivariate logistic regression models, are presented in **Table 4**. As in the bivariate analyses, hookworm infection was the strongest predictor of poor iron status for all three indicators. The strongest risk factor for anemia varied by the severity of the anemia being modeled. For example, the risk of anemia associated with hookworm infection increased as hemoglobin cutoffs for more severe anemia were used. In contrast, low serum retinol concentration was most strongly associated with mild anemia. The relative odds of anemia by any cutoff were approximately doubled with *P. vivax* malaria parasitemia, but the 95% CI included 1. In stratified analyses, malaria was strongly associated with moderate to severe anemia among the small proportion of women not infected with hookworms (25.0 versus 2.9%, $P = 0.08$ by Fisher's exact test). However, the small numbers in some of the subgroups did not allow for adequate analysis of this interaction.

Hookworm infection was most strongly associated with depleted iron stores. The adjusted odds of serum ferritin of <10 μg/L were approximately three to nine times greater for increasingly heavy hookworm infection compared with odds for women without infection. In contrast, *P. vivax* malaria parasitemia and low serum retinol were not associated with increased odds of low serum ferritin after multivariate adjustment. The AOR of iron-deficient erythropoiesis for categories of hookworm egg counts of ≥1000 eggs/g of feces was ~2 compared with women without hookworm infection, but this association was not statistically significant because the CI for the AOR included 1.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

TABLE 4

Adjusted odds ratios (AOR) for anemia, iron-deficient erythropoiesis and depleted iron stores associated with hookworms, Plasmodium vivax malaria and low serum retinol[1]

Risk factor	Anemia (low hemoglobin), *g/L*			Iron-deficient erythropoiesis[2]	Depleted iron stores[3]
	<110 g/L	<100 g/L	<90 g/L		
	AOR (95% CI)[4]				
Hookworm infection					
0 epg[5] (n = 49)	1.0	1.0	1.0	1.0	1.0
1–999 epg (n = 88)	1.82 (0.83, 3.96)	1.81 (0.76, 4.30)	1.97 (0.50, 7.70)	0.99 (0.46, 2.10)	2.75 (1.25, 6.08)
1000–1999 epg (n = 29)	2.86 (0.97, 8.47)	2.25 (0.76, 6.71)	4.71 (1.04, 21.32)	1.97 (0.70, 5.56)	3.33 (1.17, 9.45)
≥2000 epg (n = 24)	2.97 (0.93, 9.52)	7.47 (2.26, 24.68)	8.24 (1.73, 39.21)	2.35 (0.76, 7.29)	8.81 (2.54, 30.49)
P. vivax malaria parasitemia					
Negative (n = 142)	1.0	1.0	1.0	1.0	1.0
Positive (n = 36)	2.24 (0.91, 5.52)	2.01 (0.90, 4.50)	2.28 (0.86, 6.09)	1.67 (0.74, 3.76)	1.33 (0.59, 3.03)
Serum retinol					
≥1.05 μmol/L (n = 100)	1.0	1.0	1.0	1.0	1.0
<1.05 μmol/L (n = 90)	2.52 (1.28, 4.98)	1.96 (1.01, 3.83)	2.07 (0.84, 5.06)	1.69 (0.89, 3.19)	1.06 (0.55, 2.04)

[1] Analysis restricted to women who provided a stool specimen (n = 190). AOR and their SE estimated from multivariate logistic regression models that included all three risk factors and trimester of pregnancy.
[2] Iron-deficient erythropoiesis defined as erythrocyte protoporphyrin >70 μmol/mol heme.
[3] Depleted iron stores defined as serum ferritin < 10 μg/L.
[4] CI, confidence interval.
[5] epg, eggs/g of feces.

In multivariate regression models of iron status indicators as continuous variables, hookworm infection intensity remained the strongest predictor of all three iron status indicators. The bivariate associations of hookworm infection intensity with hemoglobin and serum ferritin concentrations (Table 3) were essentially unchanged by adjustment for other risk factors and maternal characteristics. Serum retinol of <1.05 μmol/L and *P. vivax* malaria parasitemia were each associated with a hemoglobin decrement of ~5 g/L after adjustment. When the hemoglobin model was run with serum retinol as a continuous variable instead, a 1-μmol/L increase in retinol was associated with a 9-g/L increase in hemoglobin ($P < 0.001$). In the multivariate model for EP, there was a trend of progressively larger increments in EP with increasing intensity of hookworm infection, and the increment associated with a hookworm egg count of ≥2000 eggs/g of feces was significant ($P < 0.05$).

Interaction between low serum retinol concentration and *P. vivax* malaria was found in both the hemoglobin and EP linear regression models (P-value for retinol × malaria interaction terms: hemoglobin, 0.06; EP, 0.07). Malaria was associated with a much larger hemoglobin decrement (−10.6 g/L, $P < 0.005$) among women with low serum retinol than among those with serum retinol of ≥1.05 μmol/L (−1.3 g/L, $P = 0.70$). Conversely, low serum retinol was associated with a larger decrement in hemoglobin (−12.3 g/L, $P < 0.01$) among women with malaria parasitemia than among uninfected women (−2.9 g/L, $P = 0.18$). The interaction of these two risk factors for EP was similar in type and magnitude to the interaction for hemoglobin.

Attributable fractions for the causes of anemia (**Table 5**) were calculated to assess their importance at a population level. Approximately 40% of all cases of anemia and 85% of all cases of moderate to severe anemia were attributable to iron deficiency, making it the most important cause of anemia identified in this population. Among the other risk factors, hookworm infection was the next most important contributor to anemia, with more than half of all cases of moderate to severe anemia attributable to hookworm infection. Vitamin A deficiency was also an important contributor to anemia at the population level, with attributable fractions of 14 and 29% for all cases of anemia and moderate to severe anemia, respectively. Although the attributable fraction for all cases of anemia was small, 16% of cases of moderate to severe anemia were attributable to *P. vivax* malaria.

DISCUSSION

Prevalence of anemia and iron deficiency. Anemia is a serious public health problem among pregnant women in the rural plains of Nepal (World Health Organization, UNICEF and UNU 1998). The observed prevalence of 73% was nearly identical to a South Asian regional anemia prevalence estimate of 75% among pregnant women, the highest in the world (World Health Organization 1992). In India, 88% of pregnant women are anemic (World Health Organization, UNICEF and UNU 1998). A survey of pregnant women in Bihar State, India (across the border from Sarlahi district, Nepal), found an anemia prevalence of 81% (Agarwal et al. 1987).

Our findings provide a population-based picture of iron status during pregnancy among rural South Asian women living in conditions of chronic malnutrition and endemic infections. Iron deficiency appeared to be the dominant cause of anemia, especially moderate to severe anemia. Eighty-five percent of cases of moderate to severe anemia were attributable to iron deficiency. However, 45% of non–iron-deficient women were anemic, suggesting that other causes of anemia are present in this population. World Bank prevalence estimates of iron deficiency in the general population were 69% for India but, surprisingly, only 24% for Nepal (Levin et al. 1993). Our estimate of 81% in pregnant women suggests that the prevalence in Nepal has been underestimated and is, at least in the terai region of the country, comparable to that of other areas of South Asia.

Our findings are indicative of progressive iron depletion during pregnancy. Hemoglobin, EP and serum ferritin concentrations were indicative of poor iron status overall and were

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

TABLE 5

Attributable fractions of anemia among Nepalese pregnant women

Risk factor/form of anemia	Prevalence of anemia in nonexposed group (p_1)	Adjusted odds ratio[1]	Adjusted prevalence ratio[2]	Proportion of cases exposed to risk factor (p_d)	Attributable fraction[3]
	%			%	
Iron deficiency[4]					
All anemia[5]	44.6	···	1.78[8]	88.1	39
Moderate to severe anemia[6]	3.1	···	7.80[8]	97.0	85
Hookworm infection					
All anemia	55.1	2.15	1.32	78.7	19
Moderate to severe anemia	8.2	3.15	2.68	86.7	54
Plasmodium vivax malaria parasitemia					
All anemia	63.4[9]	2.24	1.25	23.7	5
Moderate to severe anemia	13.4[9]	2.28	1.95	32.1	16
Low serum retinol[7]					
All anemia	58.0[9]	2.52	1.34	54.3	14
Moderate to severe anemia	11.0[9]	2.07	1.85	63.3	29

[1] Adjusted odds ratio (AOR) estimated from multivariate logistic regression models that included hookworm infection, *P. vivax* malaria parasitemia, vitamin A deficiency (serum retinol <1.05 μmol/L) and trimester of pregnancy.

[2] Adjusted prevalence ratio (APR) calculated from AOR using the following conversion formula: $APR = AOR/[1 + p_1(AOR - 1)]$, where p_1 = prevalence of anemia in the nonexposed group (Osborn and Cattaruzza 1995).

[3] Defined as $[p_d(RR - 1)/RR]$ where p_d is proportion of cases of anemia exposed to a risk factor, and RR is relative risk or APR (Kleinbaum et al. 1982, Rockhill et al. 1998).

[4] Erythrocyte protoporphyrin >70 μmol/mol heme or serum ferritin <10 μg/L.

[5] Hemoglobin <110 g/L.

[6] Hemoglobin <90 g/L.

[7] Serum retinol <1.05 μmol/L.

[8] Defined as prevalence of anemia among those with iron deficiency divided by prevalence among those without iron deficiency and not adjusted for other risk factors.

[9] Prevalence in the nonexposed group from the subsample of respondents with complete risk factor data, $n = 190$.

worst in the third trimester, suggesting that the high prevalence of anemia was caused by underlying iron deficiency. The extremely low serum ferritin concentrations observed in this study are evidence that rural Nepalese women enter pregnancy with depleted iron stores.

Risk factors for anemia and iron deficiency

Hookworm infection. We previously reported that hookworm infection is endemic among women in the rural plains of Nepal (Navitsky et al. 1998), and the present results suggest that it is responsible for 54% of cases of moderate to severe anemia during pregnancy. Hookworm infection was associated with all three indicators of iron status in a density-dependent manner. Mature hookworms cause intestinal bleeding, leading to fecal blood loss proportional to the intestinal worm burden (Roche and Layrisse 1966). The intensity of hookworm infection that causes iron deficiency anemia varies according to the species and the iron status of the population. Hookworms in this study sample were exclusively *A. duodenale* (Navitsky et al. 1998), the hookworm species that causes the greatest blood loss (Pawlowski et al. 1991), and iron deficiency was severe. Thus, hookworm infection exacerbated iron deficiency and anemia in this setting.

Hookworm infection has been established as a strong predictor of iron deficiency and anemia in other populations (Hopkins et al. 1997, Layrisse and Roche 1964, Roche and Layrisse 1966, Stoltzfus et al. 1997), but few studies have examined these relationships in pregnant women. Hookworm infection was associated with severe but not moderate anemia among women receiving antenatal care at a hospital in Kathmandu, Nepal (Bondevik et al. 2000). A Kenyan study of anemia in pregnancy reported that women with hookworm egg counts of ≥1000 eggs/g feces had a lower hemoglobin concentration than women with <1000 eggs/g feces (Shulman et al. 1996). A single course of anthelminthic therapy in addition to iron-folate supplementation significantly increased hemoglobin concentrations and improved iron status (serum ferritin and EP) in pregnant Sri Lankan plantation workers, suggesting that hookworm infection caused iron deficiency anemia in that population (Atukorala et al. 1994). However, allocation to anthelminthic therapy was nonrandom and the prevalence and intensity of hookworm infection were not assessed.

Malaria. In Sarlahi, where the prevalence of malaria parasitemia was relatively low (20%) and only *P. vivax* was identified, malaria parasitemia more than doubled the odds of moderate to severe anemia after control for other causes and was associated with anemia in both nulliparous and parous women. This is one of the few community-based studies to identify *P. vivax* malaria as a contributor to pregnancy anemia. A clinic-based study in India recently reported that pregnant women infected with *P. vivax* malaria were significantly more anemic than noninfected pregnant control subjects (Singh et al. 1999). The hemoglobin decrement reported among *P. vivax*-infected patients in that study (−10 g/L) was larger than the decrement among infected women in our study (−5 g/L, see Table 3) and may be explained in part by differences in disease severity. Malaria-infected women in the Indian study were initially identified by clinical symptoms rather than by screening all women, as was done in our cohort study, and therefore represent only the most severe, symptomatic cases.

Studies of *P. falciparum* malaria have found it to be an even stronger contributor to pregnancy anemia, particularly among

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

nulliparous women (Brabin et al. 1990, Fleming 1989, Matteelli et al. 1994, McGregor 1984, Shulman et al. 1996), suggesting that the epidemiology of malaria and anemia in pregnant women may differ by species of malaria parasite. For example, *P. vivax* parasites only infect reticulocytes (<2% of red blood cells) rather than invading red blood cells of all ages, and their erythrocytic asexual phase in the human host remains in peripheral circulation rather than entering capillaries of internal organs. These differences result in heavier parasitemia with *P. falciparum* associated with serious complications and even death (Gilles and Warrell 1993, Markell et al. 1992). However, unlike *P. falciparum* malaria, relapses of *P. vivax* malaria can occur for years after an initial attack, leading to further red cell destruction and worsening anemia (Gilles and Warrell 1993, Markell et al. 1992). South Indian men infected with *P. vivax* malaria showed a progressive decrease in hemoglobin concentration, packed cell volume and red blood cell level with increasing number of malaria attacks, and this relationship held across all levels of parasitemia (Selvam and Baskaran 1996).

Vitamin A deficiency. The prevalence of vitamin A deficiency in this sample was very high. Twenty-one percent of women had a serum retinol concentration of <0.70 μmol/L, and 54% had a serum retinol concentration of <1.05 μmol/L, virtually the same as previously reported in other concurrent studies in the same population (Christian et al. 1998b). Women with a low serum retinol concentration were more than twice as likely to be anemic compared with those with a higher serum retinol concentration, suggesting that vitamin A deficiency decreases hemoglobin synthesis. Numerous studies that examined the effect of vitamin A deficiency on iron status have found an association between serum retinol and hemoglobin concentrations in pregnant women (Bondevik et al. 2000, Suharno et al. 1992), adolescent girls (Ahmed et al. 1996) and children (Mejía et al. 1977, Mohanram et al. 1977, Wolde-Gebriel et al. 1993). Suharno et al. (1992) and Ahmed et al. (1996) found 4- to 10-g/L increases in hemoglobin associated with a 1-μmol/L increase in serum retinol concentration in multivariate linear regression models adjusted for confounders. When we adjusted for other risk factors, we found a relationship of similar magnitude between serum retinol and hemoglobin (see Results).

Randomized trials to examine the effect of vitamin A supplementation on iron status have reinforced the findings from observational studies. A single, massive dose of vitamin A significantly increased hemoglobin among vitamin A–deficient Thai children (Bloem et al. 1990) and among Indonesian children who were both vitamin A deficient and anemic (Semba et al. 1992). In a randomized, placebo-controlled trial among anemic Guatemalan children, daily vitamin A supplements for 2 mo increased hemoglobin concentration by 9 g/L compared with an increase of 3 g/L in the placebo group (Mejía and Chew 1988). The consumption of vitamin A–fortified monosodium glutamate increased the hemoglobin of Indonesian children by 10 g/L after 5 mo compared with no change among control children (Muhilal et al. 1988). A randomized, placebo-controlled trial of vitamin A and iron supplementation among pregnant Indonesian women found that 2 mo of daily vitamin A supplementation significantly increased the hemoglobin concentration and reduced the prevalence of anemia by 23% (Suharno et al. 1992). Among women who received both vitamin A and iron, the positive effect on hemoglobin was even greater, and anemia was almost completely eliminated.

We observed a significant interaction of serum retinol and *P. vivax* malaria on hemoglobin and EP. These findings suggest a synergistic negative effect of vitamin A deficiency and malaria parasitemia on erythropoiesis. However, the mechanism of this interaction is unclear. Vitamin A deficiency may inhibit erythropoiesis directly (West and Roodenburg 1992) or through inhibition of iron mobilization or transport (Bloem et al. 1989, Wolde-Gebriel et al. 1993). Malaria attacks hemolyze both infected and uninfected red cells, and the immune response of the body to infection results in further red cell destruction and progressive anemia due to phagocytosis (Markell et al. 1992).

Implications for control of anemia and iron deficiency

The very high prevalence of iron deficiency and anemia during pregnancy in this study points to an urgent need for anemia treatment and control in the plains of Nepal. Although improvement in the amount and quality of dietary iron intake among pregnant women is an important long-term goal, it is well established that this alone cannot meet the iron needs of pregnancy, particularly among populations who consume diets of low iron bioavailability. Thus, the need remains to supplement the iron intake of women during pregnancy (Stoltzfus and Dreyfuss 1998, World Health Organization, UNICEF and UNU 1998) and possibly throughout the reproductive years to build up the iron stores necessary to meet the high iron demands of pregnancy and ubiquitous infections.

Anthelminthic therapy is inexpensive and is safe during pregnancy after the first trimester (World Health Organization 1996). The World Health Organization recommends anthelminthic therapy for women to control hookworm infection in areas in which the prevalence of infection is >20–30% and anemia is prevalent (World Health Organization 1996). The more severe anemia attributable to malaria may also be reduced by antimalarial chemoprophylaxis during pregnancy, as has been demonstrated among primigravidae in several studies of *P. falciparum* malaria (Bouvier et al. 1997, Fleming et al. 1986, Gilles et al. 1969, Greenwood et al. 1989, Mutabingwa et al. 1993, Nosten et al. 1994). However, chemoprophylaxis did not affect hemoglobin concentrations in trials conducted in areas of lower malaria prevalence (Hamilton et al. 1972, Jackson and Latham 1982), and its impact is not known when only *P. vivax* malaria is present.

Finally, improving the vitamin A status of vitamin A–deficient pregnant women in addition to iron supplementation may reduce the risk of mild anemia. Vitamin A supplementation of women before or during pregnancy, or both, may be an effective intervention with multiple benefits for the health and nutritional status of women (Christian et al. 1998a, West et al. 1999). Our study was conducted as part of a randomized trial of vitamin A and β-carotene supplementation of women of childbearing age, which will allow us to examine the treatment effect of supplementation on anemia and iron deficiency. Efforts should be made to prevent pregnancy anemia and its damaging consequences using an appropriate mix of interventions that address the multiple causes of anemia and iron deficiency in the population.

ACKNOWLEDGMENTS

We thank the members of the Nepal Nutrition Intervention Project-Sarlahi (in addition to the authors) for their dedication and assistance in the implementation of this study, especially D. N. Mandal, T. R. Shakya, N. N. Acharya, G. Subedi, U. Shankar, A. Bhetwal, D. B. Khadka, B. J. Thapa and all the members of the NNIPS clinic staff. We also thank Paul Connor for his assistance with data management and Kerry Schulze, Laura Caulfield and Anu Shankar for their input and discussion of the study findings.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

LITERATURE CITED

ACC/SCN (1992) Second Report on the World Nutrition Situation. Vol. 1: Global and Regional Results. ACC/SCN, Geneva.

Agarwal, D. K., Agarwal, K. N. & Tripathi, A. M. (1987) Nutritional status in rural pregnant women of Bihar and Uttar Pradesh. Ind. Pediatr. 24: 119–125.

Agarwal, R.M.D., Tripathi, A. M. & Agarwal, K. N. (1983) Cord blood haemoglobin, iron and ferritin status in maternal anaemia. Acta Paediatr. Scand. 72: 545–548.

Ahmed, F., Khan, M. R., Karim, R., Taj, S., Hyderi, T., Faruque, M. O., Margetts, B. M. & Jackson, A. A. (1996) Serum retinol and biochemical measures of iron status in adolescent schoolgirls in urban Bangladesh. Eur. J. Clin. Nutr. 50: 346–351.

Allen, L. H. (1997) Pregnancy and iron deficiency: unresolved issues. Nutr. Rev. 55: 91–101.

Atukorala, T.M.S., de Silva, L.D.R., Dechering, W.H.J.C., Dassenaeike, T.S.C. & Perera, R. S. (1994) Evaluation of effectiveness of iron-folate supplementation and anthelminthic therapy against anemia in pregnancy: a study in the plantation sector of Sri Lanka. Am. J. Clin. Nutr. 60: 286–292.

Bloem, M. W., Wedel, M. & Egger, R. J. (1989) Iron metabolism and vitamin A deficiency in children in Northeast Thailand. Am. J. Clin. Nutr. 50: 332–338.

Bloem, M. W., Wedel, M., van Agtmaal, E. J., Speek, A. J., Saowakontha, S. & Schreurs, W.H.P. (1990) Vitamin A intervention: short-term effects of a single, oral, massive dose on iron metabolism. Am. J. Clin. Nutr. 51: 76–79.

Bondevik, G. T., Eskeland, B., Ulvik, R. J., Ulstein, M., Lie, R. T., Schneede, J. & Kvåle, G. (2000) Anaemia in pregnancy: possible causes and risk factors in Nepali women. Eur. J. Clin. Nutr. 54: 3–8.

Bothwell, T. H. & Charlton, R. W. (1984) Iron Deficiency in Women, 2nd ed. International Nutritional Anemia Consultative Group, Washington, D.C.

Bouvier, P., Doumbo, O., Breslow, N., Robert, C. F., Mauris, A., Picquet, M., Kouriba, B., Dembele, H. K., Delley, V. & Rougemont, A. (1997) Seasonality, malaria, and impact of prophylaxis in a West African village. I. Effect on anemia in pregnancy. Am. J. Trop. Med. Hyg. 56: 378–383.

Brabin, B. J., Ginny, M., Sapau, J., Galme, K. & Paino, J. (1990) Consequences of maternal anaemia on outcome of pregnancy in a malaria endemic area in Papua New Guinea. Ann. Trop. Med. Parasitol. 84: 11–24.

Christian, P., West, K. P., Jr.., Khatry, S. K., Katz, J., LeClerq, S. C., Pradhan, E. K. & Shrestha, S. R. (1998a) Vitamin A or beta-carotene supplementation reduces but does not eliminate maternal night blindness in Nepal. J. Nutr. 128: 1458–1463.

Christian, P., West, K. P., Jr.., Khatry, S. K., Katz, J., Shrestha, S. R., Pradhan, E. K., LeClerq, S. C. & Pohkrel, R. P. (1998b) Night blindness of pregnancy in rural Nepal: nutritional and health risks. Int. J. Epidemiol. 27: 231–237.

Craft, N. E. (1996) High resolution HPLC method for the simultaneous analysis of carotenoids, retinoids, and tocopherols. FASEB. J. 10: A3039 (abs.).

DeBenaze, C., Galan, P., Wainer, R. & Hercberg, S. (1989) Prevention de l'anemie ferroprive au cours de la grassesse par un supplementation martiale precoce: un essaie controle. Rev. Epidemiol. Sante Publ. 27: 109–119.

DeLeeuw, N.K.W., Lowenstein, L. & Hsieh, Y. S. (1966) Iron deficiency and hydremia in normal pregnancy. Medicine (Baltimore) 45: 291.

DeMaeyer, E. M. & Adiels-Tegman, M. (1985) The prevalence of anaemia in the world. Rapp. Trimest. Statist. Sanit. Mond. 38: 302–316.

Fleming, A. F. (1989) The aetiology of severe anaemia in pregnancy in Ndola, Zambia. Ann. Trop. Med. Parasitol. 83: 37–49.

Fleming, A. F., Ghatoura, G.B.S., Harrison, K. A., Briggs, N. D. & Dunn, D. T. (1986) The prevention of anaemia in pregnancy in primigravidae in the Guinea savanna of Nigeria. Ann. Trop. Med. Parasitol. 80: 211–233.

Garn, S. M., Ridella, S. A., Petzold, A. S. & Falkner, F. (1981) Maternal hematologic levels and pregnancy outcomes. Semin. Perinatol. 5: 155–162.

Gilles, H. M., Lawson, J. B., Sibelas, M., Voller, A. & Allan, N. (1969) Malaria, anaemia and pregnancy. Ann. Trop. Med. Parasitol. 63: 245–263.

Gilles, H. M. & Warrell, D. A. (1993) Bruce-Chwatt's Essential Malariology, 3rd ed. Edward Arnold, London.

Greenwood, B. M., Greenwood, A. M., Snow, R. W., Byass, P., Bennett, S. & N'Jie, A.B.H. (1989) The effects of malaria chemoprophylaxis given by traditional birth attendants on the course and outcome of pregnancy. Trans. Trop. Med. Hyg. 83: 589–594.

Hamilton, P. .S., Gebbie, D.A.M., Wilks, N. E. & Lothe, F. (1972) The role of malaria, folic acid deficiency and haemoglobin AS in pregnancy at Mulago hospital. Trans. Trop. Med. Hyg. 66: 594–602.

Hopkins, R. M., Gracey, M. S., Hobbs, R. P., Spargo, R. M., Yates, M. & Thompson, R.C.A. (1997) The prevalence of hookworm infection, iron deficiency and anaemia in an Aboriginal community in north-west Australia. Med. J. Aust. 166: 241–244.

Jackson, R. T. & Latham, M. C. (1982) Anaemia of pregnancy in Liberia, West Africa: a therapeutic trial. Am. J. Clin. Nutr. 35: 710–714.

Kahn, H. A. & Sempos, C. T. (1989) Statistical Methods in Epidemiology. Oxford University Press, New York.

Kaneshige, E. (1981) Serum ferritin as an assessment of iron stores and other hematologic parameters during pregnancy. Obstet. Gynecol. 57: 238–242.

Klebanoff, M. A., Shiono, P. H., Selby, J. V., Trachtenberg, A. I. & Graubard, B. I. (1991) Anemia and spontaneous preterm birth. Am. J. Obstet. Gynecol. 164: 59–63.

Kleinbaum, D. G., Kupper, L. L. & Morgenstern, H. (1982) Epidemiologic Research. Lifetime Learning Publications, Belmont, CA.

Layrisse, M. & Roche, M. (1964) The relationship between anemia and hookworm infection: results of surveys of rural Venezuelan population. Am. J. Hyg. 79: 279–301.

Levin, H. M., Pollitt, E., Galloway, R. & McGuire, J. (1993) Micronutrient deficiency disorders. In: Disease Control Priorities in Developing Countries (Jamison, D. T., Mosley, W. H., Measham, A. R. & Bobadilla, J. L., eds.), pp. 421–451. Oxford University Press Oxford, U.K.

Llewellyn-Jones, D. (1965) Severe anaemia in pregnancy. Aust. N.Z. J. Obstet. Gynaecol. 5: 191–197.

MacPhail, A. P., Charlton, R. W., Bothwell, T. H. & Torrance, J. D. (1980) The relationship between maternal and infant iron status. Scand. J. Haematol. 25: 141–150.

Markell, E. K., Voge, M. & John, D. T. (1992) Markell and Voge's Medical Parasitology, 7th ed., pp. 96–125. W. B. Saunders, Philadelphia, PA.

Matteelli, A., Donato, F., Shein, A., Muchi, J. A., Leopardi, O., Astori, L. & Carosi, G. (1994) Malaria and anaemia in pregnant women in urban Zanzibar, Tanzania. Ann. Trop. Med. Parasitol. 88: 475–483.

McGregor, I. A. (1984) Epidemiology, malaria and pregnancy. Am. J. Trop. Med. Hyg. 33: 517–525.

Mejía, L. A. & Chew, F. (1988) Hematological effect of supplementing anemic children with vitamin A alone and in combination with iron. Am. J. Clin. Nutr. 48: 595–600.

Mejía, L. A., Hodges, R. E., Arroyave, G., Viteri, F. & Torun, B. (1977) Vitamin A deficiency and anemia in Central American children. Am. J. Clin. Nutr. 30: 1175–1184.

Milman, N., Agger, A. O. & Nielsen, O. J. (1994) Iron status markers and serum erythropoietin in 120 mothers and newborn infants. Acta. Obstet. Gynecol. Scand. 73: 200–204.

Milman, N., Ibsen, K. K. & Christensen, J. M. (1987) Serum ferritin and iron status in mothers and newborn infants. Acta. Obstet. Gynecol. Scand. 66: 205–211.

Mohanram, M., Kulkarni, K. A. & Reddy, V. (1977) Hematological studies in vitamin A deficient children. Int. J. Vitam. Nutr. Res. 91: 557–562.

Muhilal, Permeisih, D., Idjradinata, Y. R., Muherdiyantiningsih & Karyadi, D. (1988) Vitamin A-fortified monosodium glutamate and health, growth, and survival of children: a controlled field trial. Am. J. Clin. Nutr. 48: 1271–1276.

Murphy, J. F., O'Riordan, J., Newcombe, R. G., Coles, E. C. & Peason, J. F. (1986) Relation of haemoglobin levels in first and second trimesters to outcome of pregnancy. Lancet 1: 992–994.

Mutabingwa, T. K., Malle, L. N., de Geus, A. & Oosting, J. (1993) Malaria chemoprophylaxis during pregnancy. 2. Its effect on maternal haemoglobin, birth weight and placental malaria. Trop. Geo. Med. 45: 49–55.

Navitsky, R. C., Dreyfuss, M. L., Shrestha, J., Khatry, S. K., Stoltzfus, R. J. & Albonico, M. (1998) *Ancylostoma duodenale* is responsible for hookworm infections among pregnant women in the rural plains of Nepal. J. Parasitol. 84: 647–651.

Nepal Malaria Eradication Organization. (1986) Nepal Multidonor Review of Malaria Control Program, March 10-April 4, 1986. Vector Biology and Control Project, Arlington, VA.

Nosten, F., ter Kuile, F., Maelankiri, L., Chongsuphajaisiddhi, T., Nopdonrattakoon, L., Tangkitchot, S., Boudreau, E., Bunnag, D. & White, N. J. (1994) Mefloquine prophylaxis prevents malaria during pregnancy: a double-blind, placebo-controlled study. J. Infect. Dis. 169: 595–603.

Osborn, J. & Cattaruzza, M. S. (1995) Odds ratio and relative risk for cross-sectional data. Int. J. Epidemiol. 24: 464–465.

Pawlowski, Z. S., Schad, G. A. & Stott, G. J. (1991) Hookworm Infection and Anaemia: Approaches to Prevention and Control. World Health Organization, Geneva.

Preziosi, P., Prual, A., Galan, P., Daouda, H., Boureima, H. & Hercberg, S. (1997) Effect of iron supplementation on the iron status of pregnant women: consequences for newborns. Am. J. Clin. Nutr. 66: 1178–1182.

Puolakka, J., Jänne, O., Pakarinen, A., Järvinen, P. A. & Vihko, R. (1980) Serum ferritin as a measure of iron stores during an after normal pregnancy with and without iron supplements. Acta. Obstet. Gynecol. Scand. Suppl. 95: 43–51.

Roche, M. & Layrisse, M. (1966) The nature and causes of "hookworm anemia." Am. J. Trop. Med. Hyg. 15: 1031–1100.

Rockhill, B., Newman, B. & Weinberg, C. (1998) Use and misuse of population attributable fractions. Am. J. Pub. Health. 88: 15–19.

Romslo, I., Haram, K., Sagen, N. & Augensen, K. (1983) Iron requirement in normal pregnancy as assessed by serum ferritin, serum transferrin saturation and erythrocyte protoporphyrin determinations. Br. J. Obstet. Gynaecol. 90: 101–107.

Scholl, T. O., Hediger, M. L., Fischer, R. L. & Schaerer, J. W. (1992) Anaemia versus iron deficiency: increased risk of preterm delivery in a prospective study. Am. J. Clin. Nutr. 55: 985–992.

Selvam, R. & Baskaran, G. (1996) Hematological impairments in recurrent *Plasmodium vivax* infected patients. Jpn. J. Med. Sci. Biol. 49: 151–165.

Semba, R. D., Muhilal, West, K. P., Jr., Winget, M., Natadisastra, G., Scott, A. & Sommer, A. (1992) Impact of vitamin A supplementation on hematological indicators of iron metabolism and protein status in children. Nutr. Res. 12: 469–478.

Shrestha, S. L., Pradhan, s., Shrestha, J. P. B., Shrestha, J. D., Rajbhamdari, Y., Shrestha, G. L., Swar, T. B., Nushin, M. K. & Reisen, W. K. (1988) Observations on anopheline and malaria ecology in the far western region of Nepal, 1986. Bull. Soc. Vector. Ecol. 13: 332–342.

Shulman, C. E., Graham, W. J., Jilo, H., Lowe, B. S., New, L., Obiero, J., Snow, R. W. & Marsh, K. (1996) Malaria is an important cause of anaemia in

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023

primigravidae: evidence from a district hospital in coastal Kenya. Trans. Trop. Med. Hyg. 90: 535–539.

Singh, N., Shukla, M. M. & Sharma, V. P. (1999) Epidemiology of malaria in pregnancy in central India. Bull. World Health Org. 77: 567–572.

Stoltzfus, R. J., Chwaya, H. M., Tielsch, J. M., Schulze, K. J., Albonico, M. & Savioli, L. (1997) Epidemiology of iron deficiency anemia in Zanzibari schoolchildren: the importance of hookworms. Am. J. Clin. Nutr. 65: 153–159.

Stoltzfus, R. J. & Dreyfuss, M. L. (1998) Guidelines for the Use of Iron Supplements to Prevent and Treat Iron Deficiency Anemia. International Nutritional Anemia Consultative Group, Washington, D.C.

Suharno, D., West, C. E., Muhilal, Logman, H.G.M., de Waart, F. G., Karyadi, D. & Hautvast, J.G.A.J. (1992) Cross-sectional study on the iron and vitamin A status of pregnant women in West Java, Indonesia. Am. J. Clin. Nutr. 56: 988–993.

United Nations (1990) Controlling iron deficiency. Report of Joint ACC/SCN Workshop on Iron Deficiency Control, June 6–8, 1990. Dublin, Ireland, Administrative Committee on Coordination, Subcommittee on Nutrition, New York.

Viteri, F. E. (1994) The consequences of iron deficiency and anemia in pregnancy. In: Allen L, King J, and Lonnerdal B eds. Nutrient regulation during pregnancy, lactation, and infant growth. 127–139. Plenum, New York.

West, C. E. & Roodenburg, A. J. C. (1992) Role of vitamin A in iron metabolism. Voeding 53: 201–205.

West, K. P., Jr, Katz, J., Khatry, S. K., LeClerq, S. C., Pradhan, E. K., Shrestha, S. R., Connor, P. B., Dali, S. M., Christian, P., Pokhrel, R. P. & Sommer, A. (1999) Double blind, cluster randomised trial of low-dose supplementation with vitamin A or beta-carotene on mortality related to pregnancy in Nepal. Br. Med. J 318: 570–575.

West, K. P., Jr., Pokhrel, R. P., Katz, J., LeClerq, S. C., Khatry, S. K., Shrestha, S. R., Pradhan, E. K., Tielsch, J. M., Pandey, M. R. & Sommer, A. (1991) Efficacy of vitamin A in reducing preschool child mortality in Nepal. Lancet 338: 67–71.

World Health Organization (1991) Basic Laboratory Methods in Medical Parasitology. World Health Organization, Geneva.

World Health Organization (1992) The Prevalence of Anaemia in Women: A Tabulation of Available Information. World Health Organization, Geneva.

World Health Organization (1994) Bench Aids in the Diagnosis of Intestinal Parasites. World Health Organization, Geneva.

World Health Organization (1996) Report of the WHO Informal Consultation on Hookworm Infection and Anaemia in Girls and Women. World Health Organization, Geneva, WHO/CTD/SIP/ 96 1.

World Health Organization, UNICEF & UNU (1998) Iron Deficiency: Indicators for Assessment and Strategies for Prevention. World Health Organization, Geneva.

Wolde-Gebriel, Z., West, C. E., Haile Gebru, Tadesse, A. S., Fisseha, T., Gabre, P., Ayana, G. & Hautvast, J. G. A. J. (1993) Interrelationship between vitamin A, iodine and iron status in school children in Shoa Region, Central Ethiopia. Br. J. Nutr. 70: 593–607.

Zerfas, A. J. (1975) The insertion tape: a new circumference tape for use in nutritional assessment. Am. J. Clin. Nutr. 28: 782–787.

Zhou, L. M., Yang, W. W., Hua, J. Z., Deng, C. Q., Tao, X. & Stoltzfus, R. J. (1998) Relation of hemoglobin measured at different times in pregnancy to preterm birth and low birth weight in Shanghai, China. Am. J. Epidemiol. 148: 998–1006.

Downloaded from https://academic.oup.com/jn/article/130/10/2527/4686110 by guest on 26 June 2023